FINAL EXAMINATION
SEMESTER III, SESSION 2021/2022

COURSE CODE	:	FSPP 0034
COURSE	:	PHYSICS 2
PROGRAMME	:	FOUNDATION UTM
DURATION	:	3 HOURS
DATE	:	26th of APRIL 2022

INSTRUCTION TO CANDIDATES:

1. ANSWER ALL QUESTIONS.

WARNING!

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This examination question consists of (6) printed pages only including this page.

PART B

Waves from one bulb are emitted independently of those from the other bulb.
- Light waves from an ordinary source such as a lightbulb undergo random phase changes
- Light from bulb is incoherent.
- For interference to occur, the light source must be monochromatic.
- Light source must be coherent

Question 4 (20 MARKS)

a) A light diffraction occurs when waves impinge on an obstacle and the waves bend behind it. Consider two lightbulbs located near to each other. There is no interference pattern can be observed on a distant screen. Explain why does this happen? Justify your answer.

(4 marks)

b) Consider a separation distance between two slits is 0.25 mm where several bright fringes can be seen on a screen with 5.5 m away as illustrated in Figure 4.

i. What is the wavelength of the light if the distance of adjacent bright interference fringe from centre is 4.0 mm? Show your working steps. 182.5 nm

ii. Calculate the distance between the second-order and the fourth-order of the bright fringes. 8.03 mm

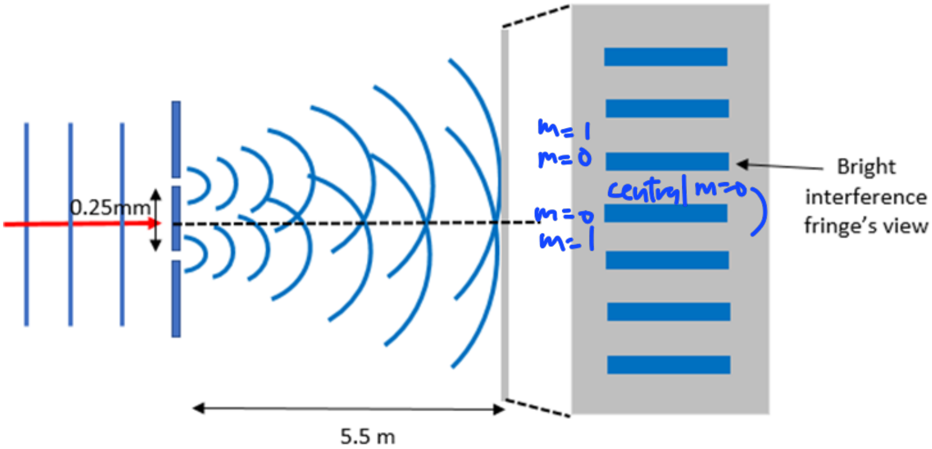

Figure 4

(12 marks)

c) Describe a blackbody radiation and how does it relate to Wien's law. Consider a star has a yellow colour with a wavelength of 60.5 nm. Calculate the possible surface temperature of the star. 479.33 mK

(4 marks)

Object that is hot (anything more than 0 K is considered "hot") emits EM radiation.
All hot objects radiate EM wave of all wavelengths.
- All incident radiation is absorbed and emits all possible radiation.
- The energy intensities of the wavelengths differ continuously from wavelength to wavelength

4) (b) $d = 0.25 \times 10^{-3}\,m$, $L = 5.5$

(i) $y_m = \frac{m\lambda L}{d}$

$4 \times 10^{-3} = \frac{(1-0)(\lambda)(5.5)}{0.25 \times 10^{-3}}$

$\lambda = 1.82 \times 10^{-7}\,m$

(ii) $y_m = \frac{(4-2)(1.82 \times 10^{-7})(5.5)}{0.25 \times 10^{-3}}$

$= 8.008 \times 10^{-3}\,m$

(c) $\lambda = \frac{0.0029}{T}$, $T = 47933\,k$

Question 5 (20 MARKS)

a) Radioactive decay is a **random** and **spontaneous** nuclear reaction. Explain the terms random and spontaneous.

(2 marks)

$T_{\frac{1}{2}} = 24 \times 60 = 1440\ s$ $\lambda = \frac{\ln(2)}{T_{\frac{1}{2}}}$

b) ^{212}Rn has a half life of 24 minutes. What does this means?

half of the atoms will have decayed in 24 min

(2 marks)

mass $214g \rightarrow 6.022 \times 10^{23}$ nuclei

$2.56 \times 10^{-6} g \rightarrow x$

$T_{\frac{1}{2}} = 1200\ s$, $T_{\frac{1}{2}} = \frac{\ln(2)}{\lambda}$

c) A laboratory has 2.56 μg of pure $^{214}_{83}Bi$, which has a half-life of 20 min.

$A = N\lambda$

i. How many nuclei are present initially? 7.2 x 10^15 nuclei

ii. What is the activity initially? 4.16 x 10^12 decay/s $A_0 = N_0 \lambda$

iii. What is the activity after 1.00 h? 5.19 x 10^11 decay/s $A = A_0 e^{-\lambda t}$

(6 marks)

d) Equation 1 shows the nuclear fission of Uranium-235 nuclei by slow moving neutrons.

$$^{235}_{92}U + {}^{1}_{0}n = {}^{139}_{54}Xe + {}^{95}_{38}Sr + 2({}^{1}_{0}n) + energy \qquad (1)$$

Calculate the energy released in the reaction. 211 MeV

$Q = +210.51\ MeV = (2) - (1)$ (exothermic)

Given:

Nuclide	Binding energy per nucleon, E/MeV
$^{95}_{38}Sr$	8.74 ×95
$^{139}_{54}Xe$	8.39 ×139
$^{235}_{92}U$	7.60 ×235

(5 marks)

Nuclei has positive charges. High temperature is required to overcome the strong
repulsive force between positive charges

e) i. Why does a fusion reaction take place at high temperature?

ii. Two deuteron fuse together to form a triton and a proton as shown in equation 2.

$$^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{1}H + {}^{1}_{1}H + Q \qquad (2)$$

Calculate the mass loss.

0.00433 u

(5 marks)

5) (a) Random => probability of a nucleus decaying at a given instant

spontaneous => cannot be predicted and independent of physical conditions and chemical changes

(c) $T_{\frac{1}{2}} = \frac{\ln(2)}{\lambda}$, $A = \frac{N}{\lambda}$

(i) $214g \Rightarrow 6.02 \times 10^{23}$

$2.56 \times 10^{-6} g \Rightarrow x$

$x = 7.2 \times 10^{15}$ nuclei

(ii) $(20 \times 60) = \frac{\ln(2)}{\lambda}$

$\lambda = 0.00058 s^{-1}$

$A = N\lambda$

$= 7.2 \times 10^{15}(0.00058)$

$= 4.18 \times 10^{12}$ decay/s

(iii) $A = A_0 e^{-\lambda t}$

$= 4.18 \times 10^{12} \cdot e^{-0.00058(3600)}$

$= 5.18 \times 10^{11}$ decays/s

APPENDIX 1

List of constant

$p = +1.60 \times 10^{-19}$ C

$e = -1.60 \times 10^{-19}$ C

$\mu_o = 4\pi \times 10^{-7}$ m A^{-1}

$\varepsilon_o = 8.85 \times 10^{-12}$ C^2 N^{-1}m^{-2}

$g = 9.80$ ms^{-2}

$c = 3 \times 10^8$ ms^{-2}

$m_e = 9.11 \times 10^{-31}$ kg

$k = 9.0 \times 10^9$ Nm^2 C^2

$h = 6.63 \times 10^{-34}$ J s

$N_A = 6.022 \times 10^{23}$

$1\ u = 1.6605 \times 10^{-27}$ kg $= 931.5$ MeV/c^2

$1\ eV = 1.60 \times 10^{-19}$ J

Refractive index of air, $n = 1.00$

Nuclide	Mass, m (u)
${}^{2}_{1}H$	2.014102
${}^{3}_{1}H$	3.016049
${}^{1}_{1}H$	1.007825

APPENDIX 2

List of Formulae

$B_o = \frac{E_o}{c}$

$a sin\theta = n\lambda$

$E = KE + W_o$

$U = \frac{1}{2}LI^2$

$\bar{P} = I_{rms}V_{rms}$

$I_{in} = I_{out}$

$R_T = R_o[1 + \alpha(T - T_o)]$

$\tau = RC$

$\tau = \frac{L}{R}$

$V_{rms} = I_{rms}R$

$V_c = V_o e^{-t/\tau}$

$I = nAve$

$M = \frac{\mu_O N_1 N_2 A}{l}$

$L = \frac{\mu_O N^2 A}{l}$

$L = \frac{N\emptyset}{I}$

$\varepsilon = \left|-L\frac{dI}{dt}\right|$

$\lambda = \frac{c}{f}$

$k = \frac{2\pi}{\lambda}$

$\omega = 2\pi f$

$m = -\frac{d_i}{d_o} = \frac{h_i}{h_o}$

$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$

$\frac{\sin\theta_i}{\sin\theta_r} = \frac{n_2}{n_1}$

$\frac{1}{f} = (n-1)\left[\frac{1}{r_1} + \frac{1}{r_2}\right]$

work function

$W_o = hf_o$

$N = N_0 e^{-\lambda t}$

$p = mv$

$KE = eV$

$T_{1/2} = \frac{\ln 2}{\lambda}$

$Q = (\Delta m)c^2$

$d sin\theta = m\lambda$ double slit bright

$m = 0,1,2,3 ....$

double slit dark

$d sin\theta = (m + \frac{1}{2})\lambda$

$m = 0,1,2,3 ...$

RC

$Q = Q_o e^{-t/\tau}$

RC

$I = I_o e^{-t/\tau}$ $\tau = RC$

time constant

$\lambda = \frac{ln2}{T_{1/2}}$

$\left|\frac{dN}{dt}\right|_o = \lambda N_o$

$A_0 = \lambda N_0$

$\left|\frac{dN}{dt}\right| = \left|\frac{dN}{dt}\right|_o e^{-\lambda t}$

$A = A_0 e^{-\lambda t}$